



Teen Mental Health & Social Media Analytics

Analysis of 1,200 teenagers aged 13-19

TOTAL PARTICIPANTS

1,200

AVG SOCIAL MEDIA

4.5h

per day

AVG SLEEP

6.5h

per night

DEPRESSION RATE

2.6%

31 cases

AVG STRESS LEVEL

5.5

out of 10

AVG ANXIETY LEVEL

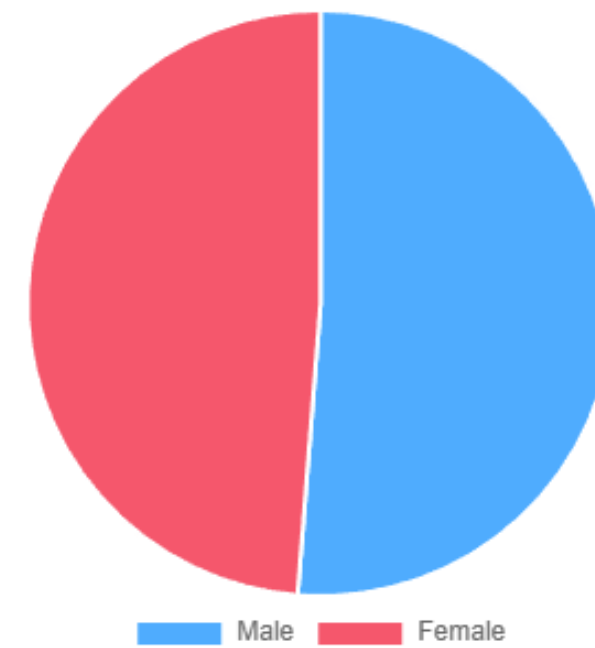
5.6

out of 10

Platform Usage Distribution

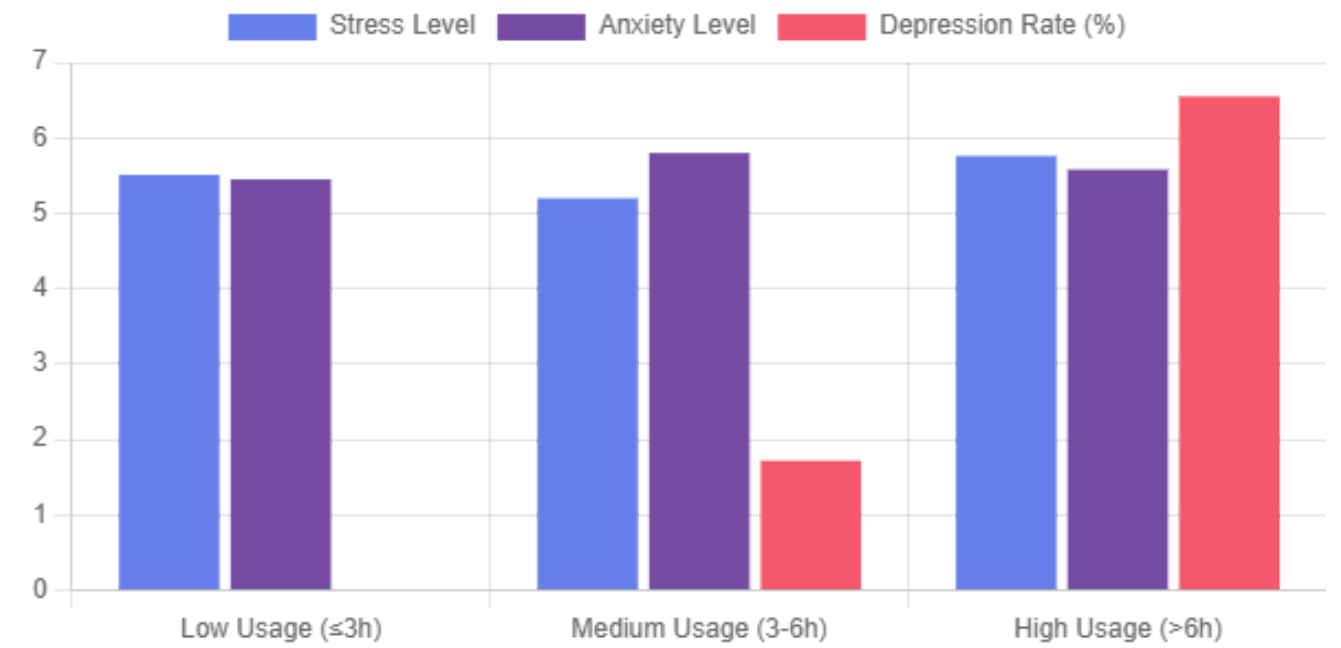
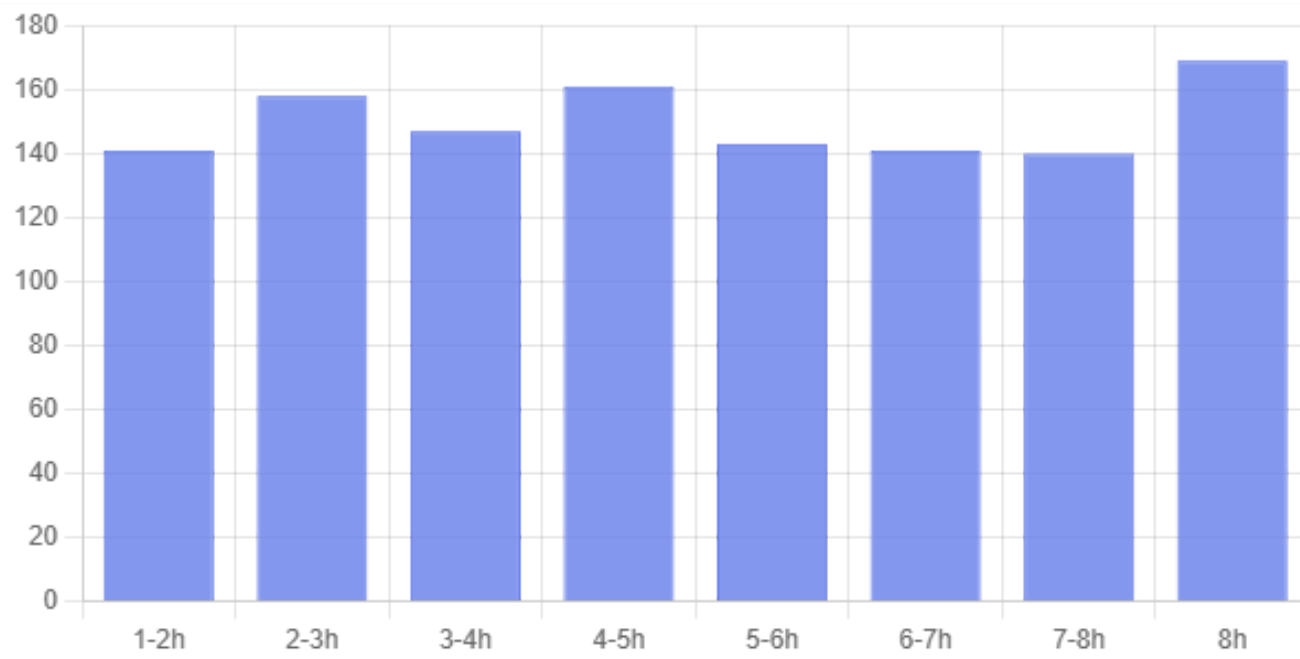


Gender Distribution

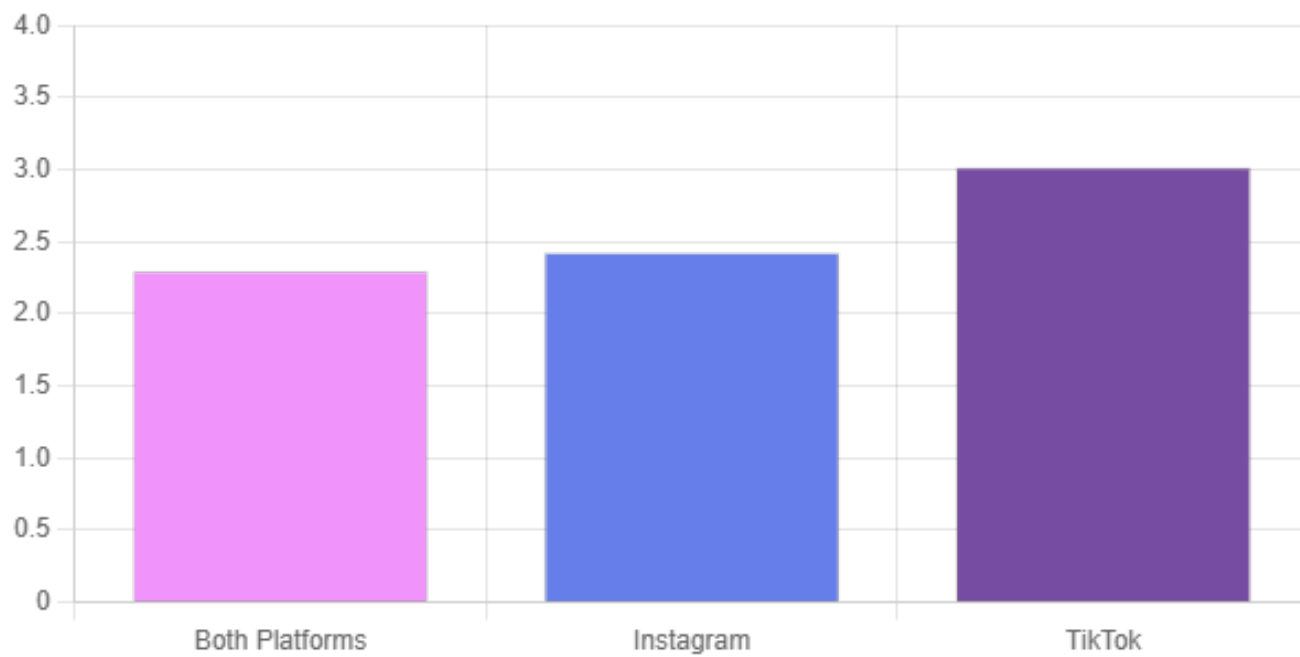


Social Media Usage Hours

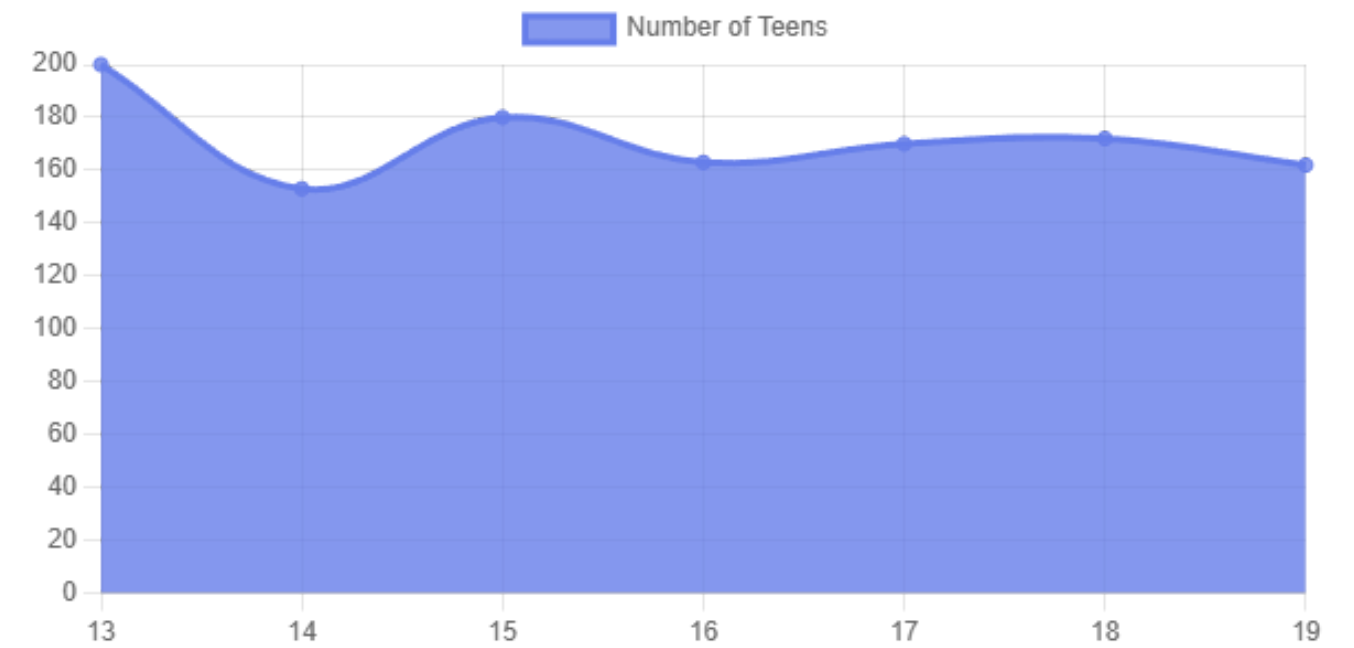
Mental Health by Usage Level



🥱 Depression Rate by Platform

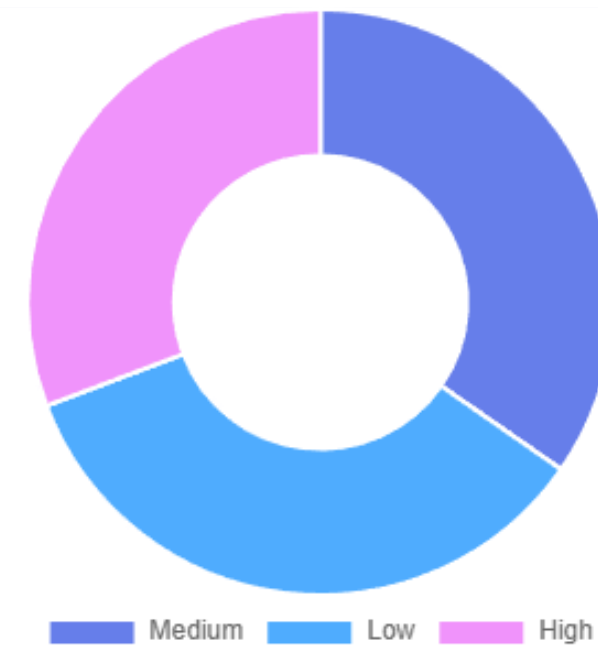
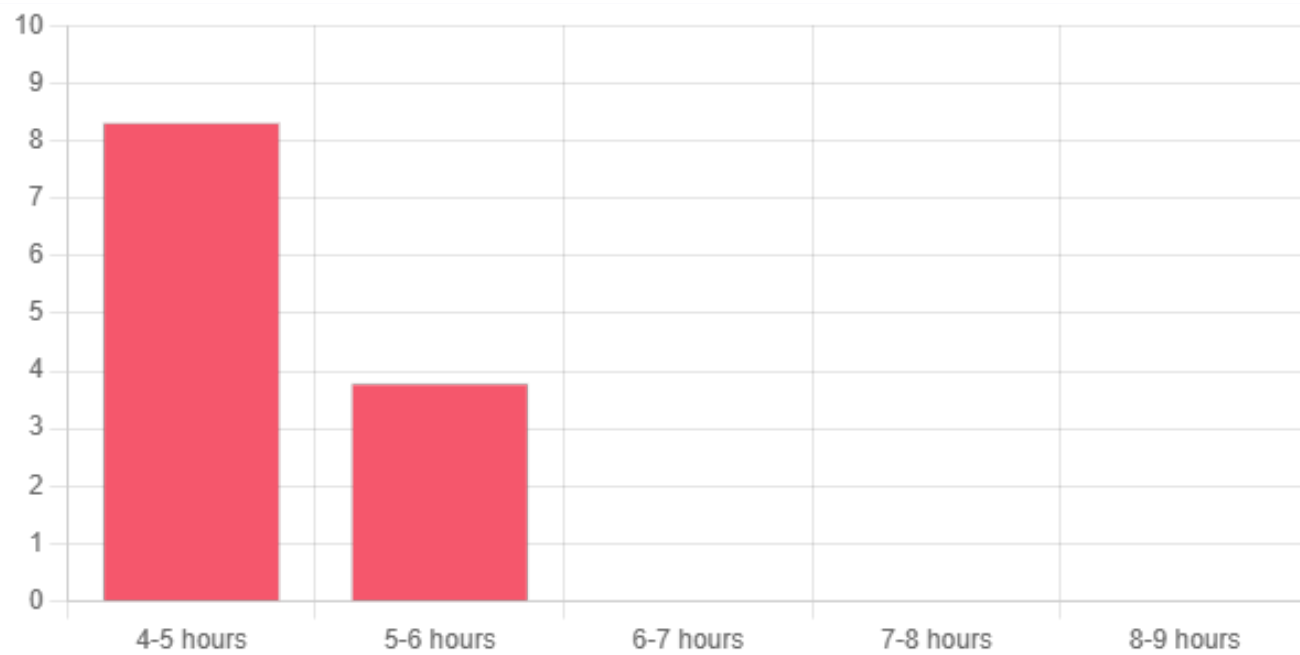


🎂 Age Distribution



😴 Sleep Hours vs Depression

💬 Social Interaction Levels



Key Correlations with Depression

Lack of Sleep (Negative)	-0.191
Social Media Hours (Positive)	+0.175
Stress Level (Positive)	+0.170
Anxiety Level (Positive)	+0.170
Physical Activity (Negative)	-0.018
Addiction Level (Negative)	-0.014
Academic Performance (Positive)	+0.001

💡 Key Insights

📱 **Heavy Social Media Usage Alert:** Teens using social media for 6+ hours daily show a **6.5% depression rate**, compared to **0% for those using 3 hours or less**.

😴 **Sleep Crisis:** Average sleep of only **6.5 hours** falls short of the recommended 8-10 hours for teenagers. Poor sleep shows a **-0.191 correlation with depression**.

📊 **Platform Patterns:** TikTok users show slightly higher depression rates (**3.0%**) compared to Instagram (**2.4%**) and Both platforms (**2.3%**).

😟 **Mental Health Triad:** Social media hours, stress, and anxiety all show similar positive correlations (~ 0.17) with depression, suggesting interconnected factors.

⚠️ **Critical Sleep Threshold:** Teens getting less than 5 hours of sleep show an **8.3% depression rate**, while those getting 6+ hours show **0% depression**.

🎯 **Low Impact Factors:** Surprisingly, addiction level (-0.014), physical activity (-0.018), and academic performance ($+0.001$) show negligible correlations with depression in this dataset.